



COURSE SLIDES

Certified Lean Six Sigma White Belt (CLSSWB) Training

Course Code: C142

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v3 · 21 July 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Certified Lean Six Sigma practitioner — process improvement, quality and data-driven problem solving.

DELIVERS

Professional short courses on Lean Six Sigma, quality management, data and cloud.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- 1 Your name and organisation / role.
- 2 A process in your work that frustrates you or your customers.
- 3 What you want to be able to recognise or improve after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

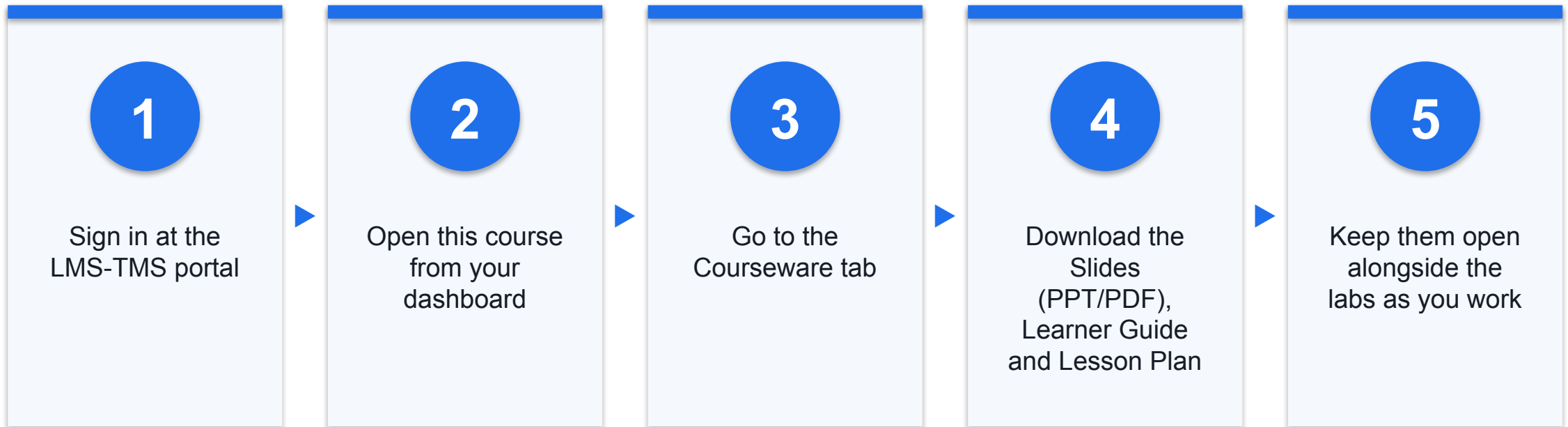
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Download Course Material



Portal: <https://lms-tms.tertiaryinfotech.com>

Lesson Plan — 1 Day, 8 hours/day

Day 1

- Day 1 — Lean Six Sigma foundations and the full DMAIC roadmap, hands-on
 - Topic 1: Lean Six Sigma Foundations
 - Topic 2: Define — Understand the Problem (Lab 1)
 - Topic 3: Measure — See What Is Really Happening (Lab 2)

Day 1 (cont.)

- Day 1 (continued)
 - Topic 4: Analyze — Find the Cause (Lab 3)
 - Topic 5: Improve — Fix the Cause (Lab 4)
 - Topic 6: Control — Hold the Gain (Lab 5)
- Daily timing
 - 9:30am–6:30pm · 1-hour lunch · tea breaks within training time

Learning Outcomes

1

Lean Six Sigma Concepts

Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.

2

The DMAIC Roadmap

Explain the five phases of the DMAIC roadmap and what happens in each.

3

Define the Problem

Identify customer requirements and a clear problem statement in the Define phase.

4

Measure & the 8 Wastes

Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.

5

Find the Root Cause

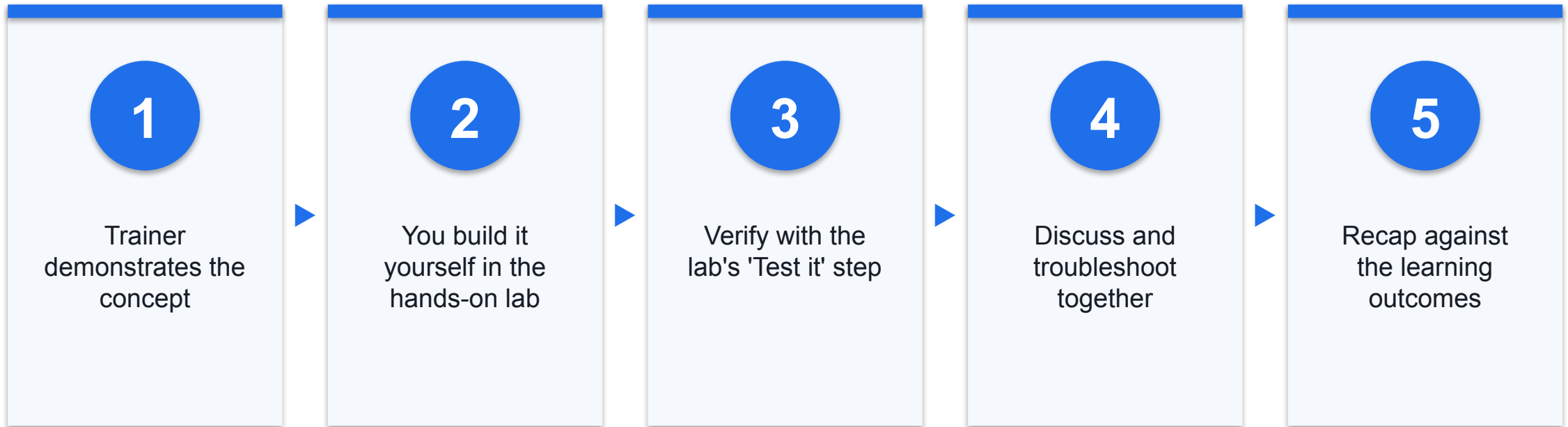
Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.

6

Improve & Control

Describe common improvement and control actions that fix a cause and hold the gain.

How You'll Learn





CORE CONCEPTS

Lean Six Sigma Fundamentals

What is Lean Six Sigma?

1

Lean removes waste

Maximise customer value by eliminating the effort the customer would never pay for.

2

Six Sigma reduces variation

A data-driven method that makes results consistent and predictable.

3

Together: fast and accurate

Lean Six Sigma delivers faster flow AND fewer defects, driven by evidence.

4

The White Belt contributes

You support an improvement team — recognise waste, speak the language, use the basic tools.

The DMAIC Roadmap

1

Define

Understand the problem — VOC, CTQ, problem statement, SMART goal and charter.

2

Measure

See what is really happening — process map, SIPOC, check sheets and the eight wastes.

3

Analyze

Find the cause — 5 Whys, Fishbone, Pareto and variation.

4

Improve

Fix the cause — countermeasures, 5S, mistake proofing, standard work and piloting.

5

Control

Hold the gain — control plan, visual management, SOPs, huddles and handover.

THE LEAN SIX SIGMA MINDSET

Improve the process, not the people.

Every tool in this course points at the process: see what it actually does, find why it does it, and change the system so the improvement holds.

One Scenario, One Complete Story

Define & Measure

- BrewBean Cafe morning rush
- VOC → CTQ and problem statement
- Process map and waste tally

Analyze

- 5 Whys chain
- Fishbone (5M) diagram
- Reading a Pareto chart

Improve & Control

- Impact/effort screening
- Pilot plan and standard work
- Control plan and handover

How Every Lab Progresses

1

Every lab uses one continuous scenario — the BrewBean Cafe morning rush, where customers queue up to 15 minutes.

2

Exactly one lab per DMAIC phase, so you walk the full roadmap in a single day.

3

Each lab produces a real artifact: a table, a map, a chart or a plan — using supplied templates.

4

Each lab ends with a 'Check your work' step so you can verify your own output.

5

By the end your five outputs form one complete improvement story, from problem to handover.

TOPIC 01

Lean Six Sigma Foundations

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

Key Concepts — Lean Six Sigma Foundations

1

What is Quality

Quality is meeting the customer's requirements — not just being free of defects.

2

What is Lean

Lean removes waste so work flows faster to the customer.

3

What is Six Sigma

Six Sigma reduces variation so results become consistent and predictable.

4

Lean Six Sigma

The two combined: faster flow AND fewer defects, decided by data not opinion.

5

Belt roles

White, Yellow, Green, Black and Master Black Belt — who does what on a team.

6

The DMAIC roadmap

Define, Measure, Analyze, Improve, Control — the five-step improvement path.

Recap — Lean Six Sigma Foundations

TOPIC 02

Define — Understand the Problem

Voice of the Customer · CTQ · Problem statement · SMART goal · Project charter

Key Concepts — Define — Understand the Problem

1

Voice of the Customer

Ask customers what they need, and record it in their own words.

2

Critical to Quality

Turn each customer need into something specific you can measure.

3

Problem statement

What is wrong, where, since when, and how big — never the solution.

4

SMART goal

Specific, Measurable, Achievable, Relevant and Time-bound.

5

Project charter

A one-page summary of the problem, goal, scope and team.

6

Scope

Agreeing what is in and out keeps a small project small.

Hands-On Labs — Define — Understand the Problem

Lab 1

- Define — Customer Requirements and the Problem Statement

Define — Customer Requirements and the Problem Statement

LAB 1

Start the BrewBean Cafe improvement the way every real project starts: find out what the customer actually wants, turn that into something measurable, and state the problem clearly enough that everyone agrees on it. The most common reason improvement projects fail is that the team never agreed what the problem was.

You'll build

A VOC-to-CTQ table, a written problem statement, a SMART goal and a simple scope table.

Tools: Voice of the Customer, CTQ, problem statement, SMART goal, project scope

Define — Customer Requirements and the Problem Statement

STEP 1 OF 6

1

Read the BrewBean Cafe scenario and underline every customer complaint you can find.

Define — Customer Requirements and the Problem Statement

STEP 2 OF 6

2

Write down five Voice of the Customer statements in the customer's own words (e.g. 'I'm late for work because the queue is too slow').

Define — Customer Requirements and the Problem Statement

STEP 3 OF 6

3

Turn each VOC statement into a measurable CTQ requirement — add a number and a unit (e.g. 'served within 5 minutes').

Define — Customer Requirements and the Problem Statement

STEP 4 OF 6



4

Write the problem statement using the template: what is wrong, where, since when, and how big. Do NOT write a solution.

Define — Customer Requirements and the Problem Statement

STEP 5 OF 6

5

Check your problem statement against the SMART test and rewrite the goal so it is Specific, Measurable, Achievable, Relevant and Time-bound.

Define — Customer Requirements and the Problem Statement

STEP 6 OF 6



6

Fill the in-scope / out-of-scope table so the team knows where the project stops.

Define — Customer Requirements and the Problem Statement

Test it

Every CTQ has a number and a unit, your problem statement contains no solution, and your goal has a metric, a baseline, a target and a date.

Recap — Define — Understand the Problem

1

You can now: Identify customer requirements and write a clear problem statement.

TOPIC 03

Measure — See What Is Really Happening

Process mapping · SIPOC · Types of data · Data collection · The eight wastes

Key Concepts — Measure — See What Is Really Happening

1

Process mapping

Draw the steps as they really happen, not as the manual says.

2

SIPOC

A one-page overview: Suppliers, Inputs, Process, Outputs, Customers.

3

Types of data

Discrete data you count; continuous data you measure on a scale.

4

Data collection

Agree what to record, who records it, and when — before you start.

5

Check sheets

A simple tally form is the easiest reliable way to collect data.

6

The eight wastes

DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

Hands-On Labs — Measure — See What Is Really Happening

Lab 2

- Measure — Map the Process and Spot the Waste

Measure — Map the Process and Spot the Waste

LAB 2

You cannot improve what you cannot see. Walk the BrewBean Cafe morning rush from the customer joining the queue to the drink being handed over, draw it as it really happens, and tally where time and effort are being lost. This is the phase that turns opinions about the problem into facts about the problem.

You'll build

A SIPOC overview, a step-by-step process map with timings, and a completed waste tally sheet.

Tools: SIPOC, process mapping, types of data, check sheets, the eight wastes (DOWNTIME)

Measure — Map the Process and Spot the Waste

STEP 1 OF 7



1

Agree the start and stop points of the process — start: customer joins the queue; stop: customer receives the drink.

Measure — Map the Process and Spot the Waste

STEP 2 OF 7

A green circle with a white number 2 inside, indicating the second step of the process.

Complete the SIPOC template: Suppliers, Inputs, Process (5-7 steps only), Outputs, Customers.

Measure — Map the Process and Spot the Waste

STEP 3 OF 7

3

List the process steps in order and write the observed time next to each one.

Measure — Map the Process and Spot the Waste

STEP 4 OF 7



4

Mark each step as value-adding (the customer would pay for it) or not value-adding.

Measure — Map the Process and Spot the Waste

STEP 5 OF 7

5

Classify the data you collected: which figures are discrete (counted) and which are continuous (measured)?

Measure — Map the Process and Spot the Waste

STEP 6 OF 7



6

Walk the process again with the DOWNTIME check sheet and tally every waste you observe against its type.

Measure — Map the Process and Spot the Waste

STEP 7 OF 7



Identify which single waste type you tallied most often, and which process step it happens at.

Measure — Map the Process and Spot the Waste

Test it

Your SIPOC has all five columns filled, every process step has a time, and every waste on your tally sheet is tagged to one of the eight DOWNTIME types.

Recap — Measure — See What Is Really Happening

1

You can now: Describe how the process runs and identify the eight wastes in it.

TOPIC 04

Analyze — Find the Cause

Root cause · 5 Whys · Fishbone diagram · Pareto chart · Variation

Key Concepts — Analyze — Find the Cause

1

Symptom vs cause

Treating the symptom makes the problem come back; treating the cause does not.

2

Root cause

The cause that, once removed, stops the problem recurring.

3

5 Whys

Keep asking 'why?' until you reach something you can actually act on.

4

Fishbone diagram

Sort possible causes into Manpower, Method, Machine, Material and Measurement.

5

Pareto chart

Roughly 80% of the problem usually comes from 20% of the causes.

6

Variation

Some variation is normal in the process; some has a specific assignable cause.

Hands-On Labs — Analyze — Find the Cause

Lab 3

- Analyze — Find the Root Cause

Analyze — Find the Root Cause

LAB 3

You now know WHERE the delay happens. This lab asks WHY it happens. Most teams jump straight from a symptom to a solution and end up fixing the wrong thing — the queue is long, so hire more staff — when the real cause was something else entirely. Here you drill past the symptom to a cause you can actually act on.

You'll build

A completed 5 Whys chain, a Fishbone diagram with causes sorted by category, and a shortlist of likely root causes.

Tools: Symptom vs cause, 5 Whys, Fishbone (5M) diagram, Pareto chart, common vs special cause variation

Analyze — Find the Root Cause

STEP 1 OF 7

1

Write the symptom from your Lab 2 findings at the top of the 5 Whys template (e.g. 'customers wait 15 minutes').

Analyze — Find the Root Cause

STEP 2 OF 7



2

Ask 'why does that happen?' five times, each answer becoming the next question. Stop when you reach something you can act on.

Analyze — Find the Root Cause

STEP 3 OF 7

3

Draw the Fishbone diagram and write the problem in the head of the fish.

Analyze — Find the Root Cause

STEP 4 OF 7

4

Brainstorm possible causes onto the five bones: Manpower, Method, Machine, Material and Measurement.

Analyze — Find the Root Cause

STEP 5 OF 7

5

Read the supplied Pareto chart of BrewBean complaint types and state which few causes account for most of the problem.

Analyze — Find the Root Cause

STEP 6 OF 7



6

Decide for each shortlisted cause whether it is common cause (happens every day) or special cause (a one-off event).

Analyze — Find the Root Cause

STEP 7 OF 7



Shortlist the two or three causes best supported by the data you collected in Lab 2.

Analyze — Find the Root Cause

Test it

Your 5 Whys chain ends in something the team can actually act on, every Fishbone cause sits under one of the five categories, and each shortlisted cause is backed by an observation, not an opinion.

Recap — Analyze — Find the Cause

1

You can now: Identify likely causes using 5 Whys and Fishbone analysis.

TOPIC 05

Improve — Fix the Cause

Generating solutions · 5S · Mistake proofing · Standard work · Piloting

Key Concepts — Improve — Fix the Cause

1

Generating solutions

Brainstorm against the proven cause — never against the symptom.

2

Choosing a solution

Compare ideas on impact, effort and risk before committing.

3

5S

Sort, Set in order, Shine, Standardise, Sustain — a tidy workplace for physical or digital work.

4

Mistake proofing

Poka-Yoke makes the error hard or impossible to make.

5

Standard work

Write the better method down so everyone does it the same way.

6

Piloting

Try the change small first, so mistakes stay cheap.

Hands-On Labs — Improve — Fix the Cause

Lab 4

- Improve — Choose and Pilot a Countermeasure

Improve — Choose and Pilot a Countermeasure

LAB 4

A cause is only worth finding if something changes because of it. Take the root causes you shortlisted in Lab 3, generate countermeasures against them, and pick one that is small enough to try next week. The discipline here is to solve the CAUSE you proved, not the symptom you started with.

You'll build

A brainstormed countermeasure list, an impact/effort grid, one selected countermeasure and a simple pilot plan.

Tools: Brainstorming, impact/effort screening, 5S, mistake proofing (Poka-Yoke), standard work, piloting

Improve — Choose and Pilot a Countermeasure

STEP 1 OF 7

1

Write your top root cause from Lab 3 at the top of the sheet — every idea must address THIS cause.

Improve — Choose and Pilot a Countermeasure

STEP 2 OF 7

2

Brainstorm at least six countermeasures without judging any of them yet.

Improve — Choose and Pilot a Countermeasure

STEP 3 OF 7

3

Check whether 5S would help: is anything at the counter hard to find, out of place or untidy?

Improve — Choose and Pilot a Countermeasure

STEP 4 OF 7

4

Check whether mistake proofing would help: could the wrong order be made impossible rather than merely discouraged?

Improve — Choose and Pilot a Countermeasure

STEP 5 OF 7

5

Plot every idea on the impact/effort grid and pick one from the high-impact, low-effort corner.

Improve — Choose and Pilot a Countermeasure

STEP 6 OF 7

6

Write the improved method as standard work — short numbered steps anyone on shift could follow.

Improve — Choose and Pilot a Countermeasure

STEP 7 OF 7

7

Write a one-week pilot plan: what changes, who runs it, what you will measure, and how you will know it worked.

Improve — Choose and Pilot a Countermeasure

Test it

Your selected countermeasure clearly addresses the root cause from Lab 3 (not the symptom), and your pilot plan names a measure, an owner and a review date.

Recap — Improve — Fix the Cause

1

You can now: Describe improvement actions that address the identified cause.

TOPIC 06

Control — Hold the Gain

Control plan · Visual management · SOPs · Team huddles · Handover

Key Concepts — Control — Hold the Gain

1

Why Control matters

Without Control, processes quietly drift back to the old way.

2

Control plan

What we measure, the target, how often, who owns it, and what to do if it slips.

3

Visual management

Make performance visible so problems are noticed the same day.

4

Standard operating procedures

Written instructions that lock in the improved method.

5

Team huddles

Short regular stand-ups catch problems while they are still small.

6

Handover

Give the improved process back to the people who run it every day.

Hands-On Labs — Control — Hold the Gain

Lab 5

- Control — Hold the Gain and Hand Over

Control — Hold the Gain and Hand Over

LAB 5

Most improvements work for a fortnight and then quietly disappear. Control is what stops that happening. Close the BrewBean Cafe story by deciding what will be measured from now on, who owns it, what happens when it slips, and how the improved process is handed back to the people who run it every day.

You'll build

A completed control plan, a visual management board sketch, and a one-page summary of the whole DMAIC story.

Tools: Control plan, visual management, SOPs, team huddles, handover, one-page summary

Control — Hold the Gain and Hand Over

STEP 1 OF 7



1

Choose the ONE measure that tells you the improvement is still working (e.g. average wait time at 8:00am).

Control — Hold the Gain and Hand Over

STEP 2 OF 7

A green circle with a white number 2 inside, indicating the second step of the process.

Complete the control plan template: measure, target, how often it is checked, and who owns it.

Control — Hold the Gain and Hand Over

STEP 3 OF 7

3

Write the reaction plan — the exact steps to take when the measure misses target.

Control — Hold the Gain and Hand Over

STEP 4 OF 7

4

Sketch a simple visual board that would make this measure visible to the whole shift.

Control — Hold the Gain and Hand Over

STEP 5 OF 7



5

Turn your Lab 4 standard work into a short SOP the cafe could actually pin up.

Control — Hold the Gain and Hand Over

STEP 6 OF 7

6

Plan a five-minute daily huddle: who attends, what is reviewed and how long it lasts.

Control — Hold the Gain and Hand Over

STEP 7 OF 7

7

Summarise the whole project on one page — problem, cause, countermeasure, result and control — and name who you hand it over to.

Control — Hold the Gain and Hand Over

Test it

Your control plan names a measure, a target, a frequency and an owner; your reaction plan says what to DO when it slips; and your one-page summary tells the complete DMAIC story from problem to handover.

Recap — Control — Hold the Gain

- 1 You can now: Describe the control actions that sustain an improvement.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Lean Six Sigma Concepts

Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.

2

The DMAIC Roadmap

Explain the five phases of the DMAIC roadmap and what happens in each.

3

Define the Problem

Identify customer requirements and a clear problem statement in the Define phase.

4

Measure & the 8 Wastes

Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.

5

Find the Root Cause

Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.

6

Improve & Control

Describe common improvement and control actions that fix a cause and hold the gain.

Continuing Your Lean Six Sigma Journey

1

Spot the eight wastes in your own workplace this week — the DOWNTIME checklist works anywhere.

2

Contribute to an improvement team using the language and tools from this course.

3

Progress towards Yellow Belt — deeper data collection, process metrics and analysis tools.

4

Build the habit: ask 'why?' past the symptom, and check the gain held after any change.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

IMPROVE THE PROCESS

Thank You!

You can now describe the DMAIC roadmap, recognise waste and variation, and contribute to a Lean Six Sigma improvement team.